

Problem 26.13

An annuity pays 10 at the end of the first year, 20 at the end of the second year, 30 at the end of the third year, etc. The last payment is made at the end of the 12th year. Calculate the accumulated value of this annuity immediately after the last payment using an annual effective rate of 4%.

Solution

The payments are

$$10, 20, 30, \dots, 120.$$

Therefore, the accumulated value at time 12 is

$$AV = 10(Is)_{\overline{12}|}.$$

For an increasing annuity-immediate accumulated immediately after the last payment,

$$(Is)_{\overline{n}|} = \frac{\ddot{s}_{\overline{n}|} - n}{i}.$$

Thus,

$$AV = 10 \left(\frac{\ddot{s}_{\overline{12}|} - 12}{0.04} \right).$$

Now,

$$s_{\overline{12}|} = \frac{(1.04)^{12} - 1}{0.04},$$

so

$$\ddot{s}_{\overline{12}|} = (1.04)s_{\overline{12}|} = 1.04 \left(\frac{(1.04)^{12} - 1}{0.04} \right) = 15.62683768.$$

Therefore,

$$AV = 10 \left(\frac{15.62683768 - 12}{0.04} \right) = 906.7094207.$$

Hence, the accumulated value is

$$\boxed{906.71}.$$